Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-I Business Mathematics Examination May 2017

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC-I) Business Mathematics. It contains comments on candidates' responses to the 2017 HSSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E- Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations:

Generally it is noted that weaker candidates are not well-versed with the hierarchy of arithmetical and algebraic operations. This is generally impeding their performance in overall paper of business mathematics.

Constructed Response Questions (CRQs)

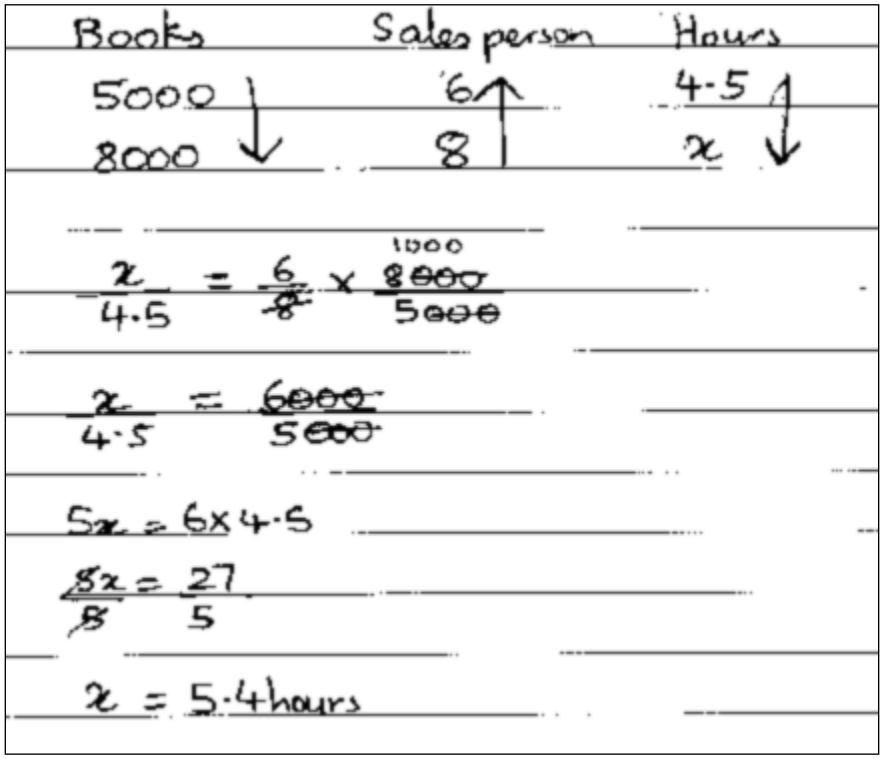
Detailed Comments:

The time required to process an order of books at a bookshop varies directly with the number of books purchased and inversely with the number of salespersons allocated to fulfil the order.

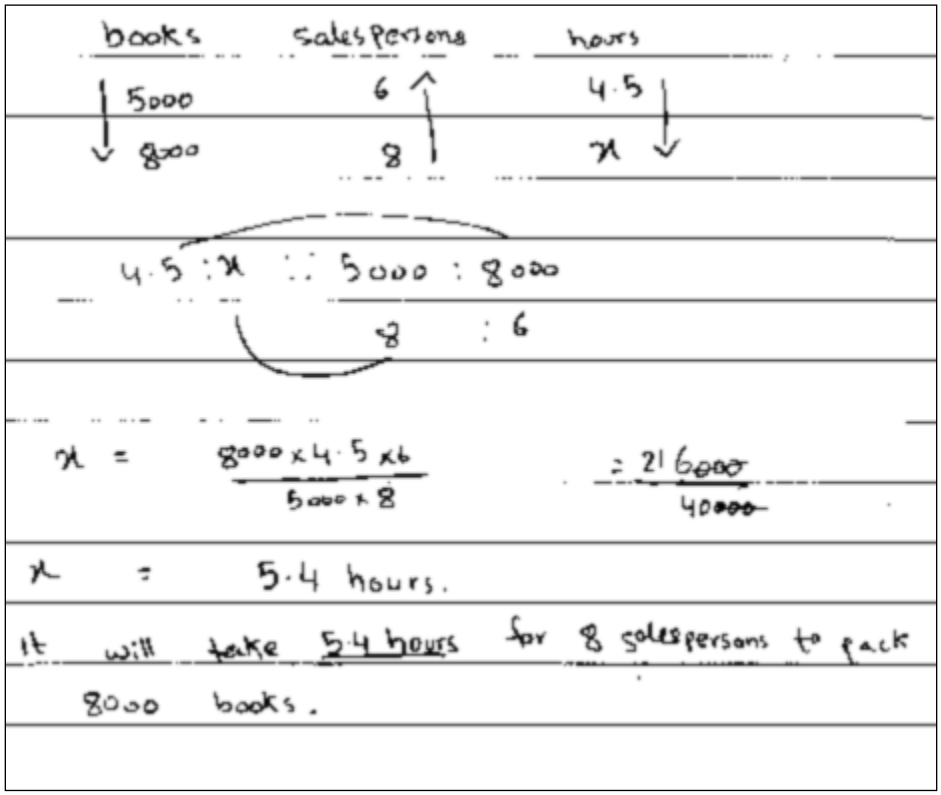
If 5,000 books can be packed by 6 salespersons in 4.5 hours, then how long would it take 8 salespersons to pack 8,000 books?

Better responses indicated that the candidates correctly established the relation between the given quantities, i.e. books, sales persons; and time required to process the order and converted it into to a correct mathematical equation $t = k\frac{b}{s}$ or $\frac{x}{4.5} = \frac{6}{8} \times \frac{8000}{5000}$ as per the information given in the question. Finally, candidates were able to simplify correctly to find the time to complete the order.

Example 1:

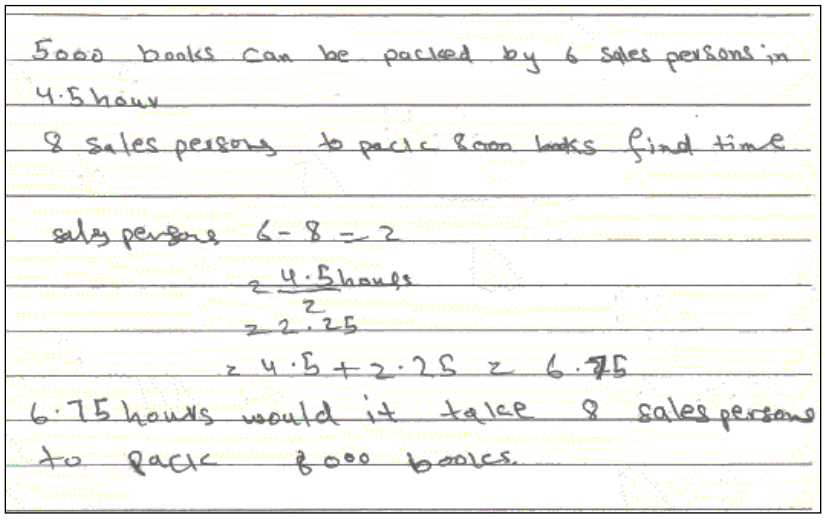


Example 2:

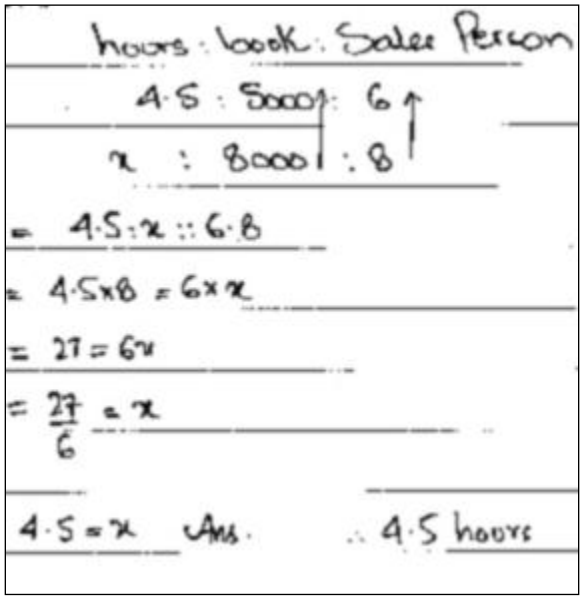


Weaker responses revealed that candidates lacked understanding about the direct or inverse proportion and, hence, failed to translate the given information mathematically. Candidates made mistakes in writing relation, e.g. $4.5\times 8 = 6\times 8$ or instead of developing direct relation, they wrote inverse relation or vice versa between the quantities and failed to solve the given problem. Following are two examples citing the misconceptions exhibited by weaker candidates.

Example 1:



Example 2:



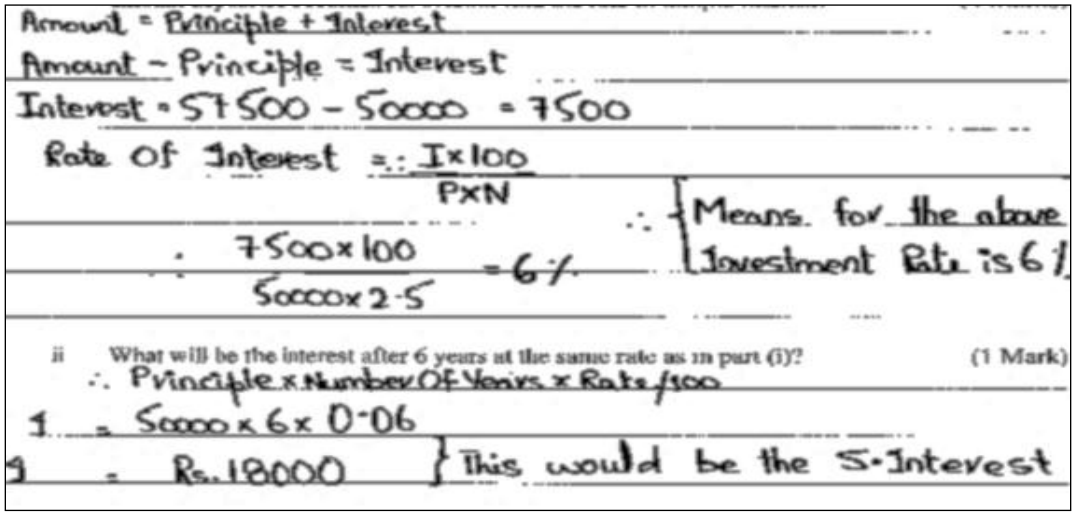
Question 2a:

i. Amina deposited Rs 50,000 in a saving scheme on simple interest. After 30 months, the amount deposited becomes Rs 57,500, find the rate of simple interest.

ii. What will be the interest after 6 years at the same rate as in part (i)?

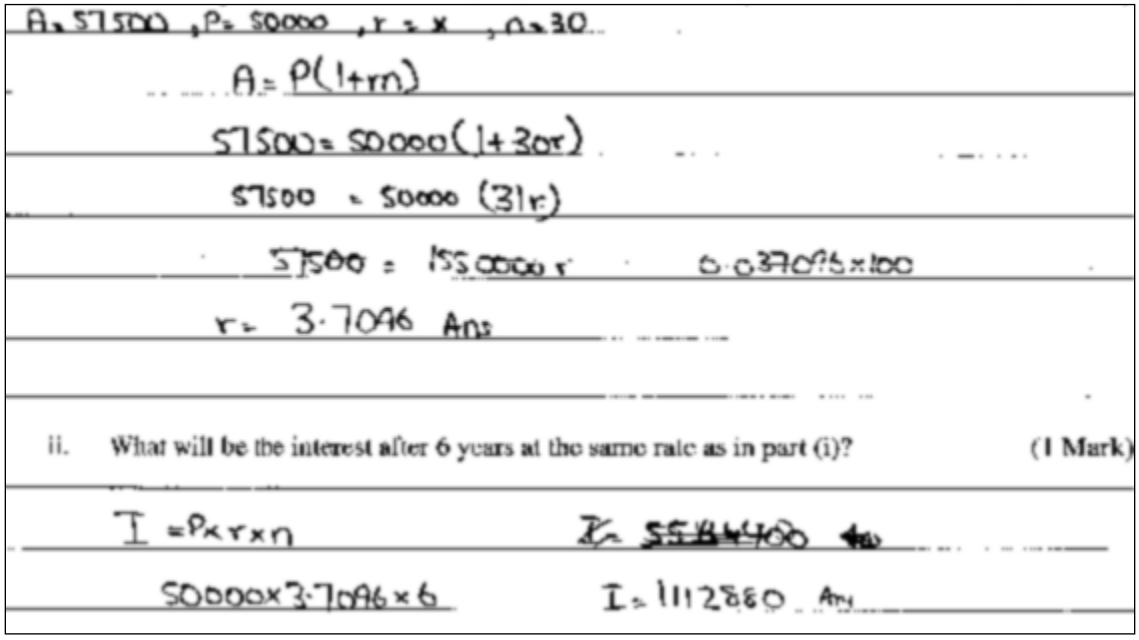
Better responses informed that the candidates applied their knowledge to calculate the interest rate based on the concept of simple interest. They calculated simple interest by subtracting original amount from the accumulated amount. These responses were also free from calculation errors and successfully found the percentage rate and interest after 6 years.

Example:



Weaker responses indicated that candidates were unable to differentiate between formula of simple interest and compound interest. Some weaker responses reported that candidates failed to find the required interest rate and interest after 6 years because they made calculation errors. In other responses, it was noted that candidates started with the assumption of interest rate to solve the problem which is an incorrect approach and can be seen in example 2.

Example 1:


A = 57500, P = 50000, r = x, n = 30

A = P(1+rn)

57500 = 50000(1+30r)

57500 = 50000 (31r)

57500 = 1550000 r 0.037096 × 100

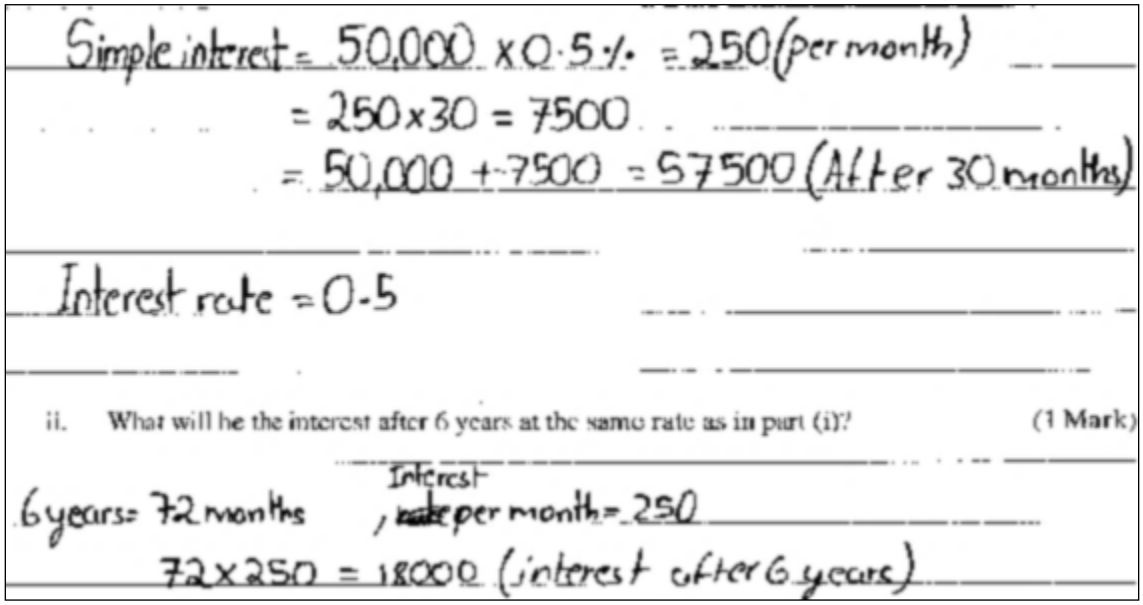
r = 3.7096 Ans

ii. What will be the interest after 6 years at the same rate as in part (i)? (1 Mark)

I = P × r × n I = 5584400 ×

50000 × 3.7096 × 6 I = 1112880 Ans

Example 2:


Simple interest = 50,000 × 0.5% = 250 (per month)

= 250 × 30 = 7500

= 50,000 + 7500 = 57500 (After 30 months)

Interest rate = 0.5

ii. What will be the interest after 6 years at the same rate as in part (i)? (1 Mark)

6 years = 72 months, Interest rate per month = 250

72 × 250 = 18000 (interest after 6 years)

Question 2b:

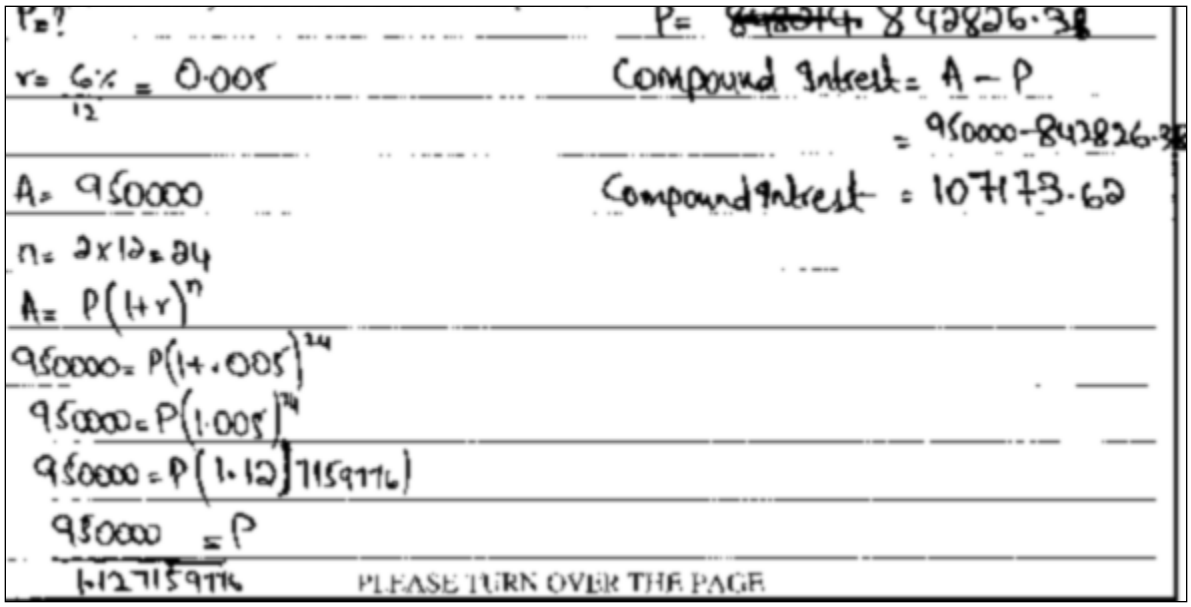
Sajjad is planning to buy a car of Rs 950,000 after two years. For this purpose, he invested in a saving scheme which pays 6% interest compounded monthly. How much amount should Sajjad invest in the scheme today so that he would have Rs 950,000 after two years? Also find the compound interest earned by Sajjad during the two years.

(**Note**: Write your answer to two decimal places)

This was generally a less attempted question with many blank scripts which was indicative of the fact that the candidates were more comfortable with the concept of simple interest, than the concept of compound interest.

Better Responses exhibited that candidate systematically found the values of *i* and number of compounding periods *n* and applied formula of compound interest correctly to find the quantities asked in the question.

Example 1:



Weaker responses indicated that candidates were unable to differentiate between principal amount and accumulated amount and, therefore, failed to apply the formula correctly. Some candidates also made calculation errors while finding compounding periods and interest rate on monthly basis. Two such responses are shown below:

Example 1:

$$P\left(1+\frac{r}{m}\right)^{mxt}$$

$$950{,}000\left(1+\frac{0.06}{12}\right)^{12X1}$$

$$1070801.787$$

$$950{,}000 - 1070801.787$$

$$120801.787 \text{ Ans.}$$

Example 2:

$A = P(1+r)^n$

$A = 950{,}000$

$r = 6\% = \frac{0.06}{6} = 0.01$

$n = 2 \times 6 = 12$

$n = 6$

$$950{,}000 = P(1+0.01)^{12}$$

$$950{,}000 = P(1.126)$$

$$\frac{950{,}000}{1.126} = P$$

$$P = ₱843694.4938 \text{ Ans.}$$

Question 3a:

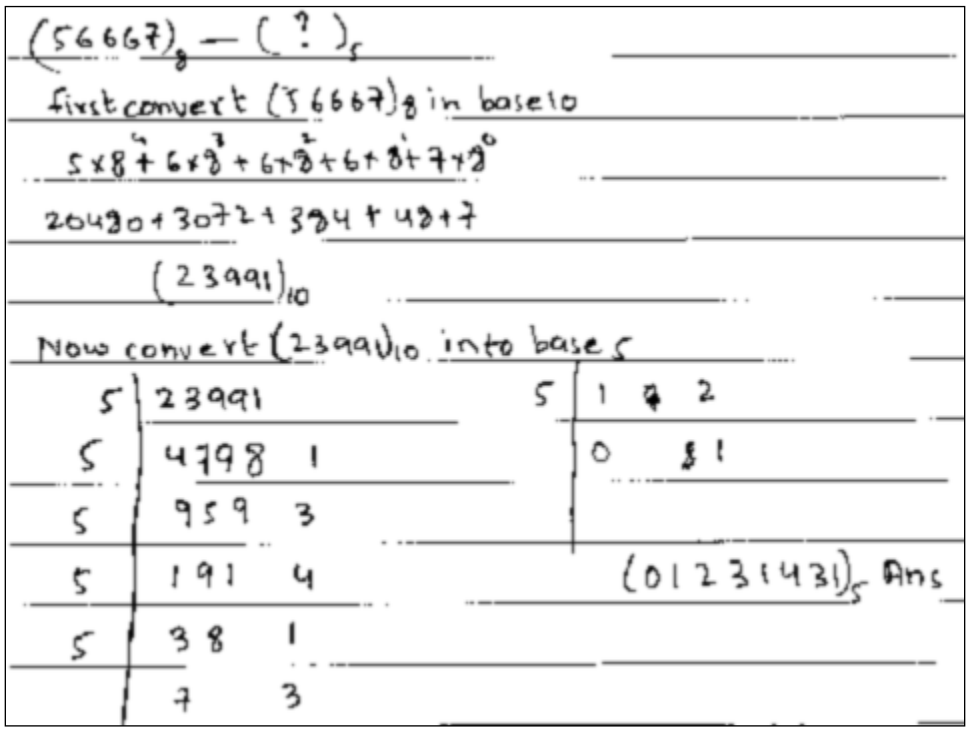
Convert $(56667)_8$ to quinary number (number in base 5).

(**Note**: Write all necessary steps

This was generally a well attempted question.

Better responses reported that the candidates were able to covert the given number in base 8 to number in base 10 (decimal system) and then correctly converted the obtained number in decimal system to base 5. They supported their answers with orderly working which demonstrated a clear understanding of the concept being examined.

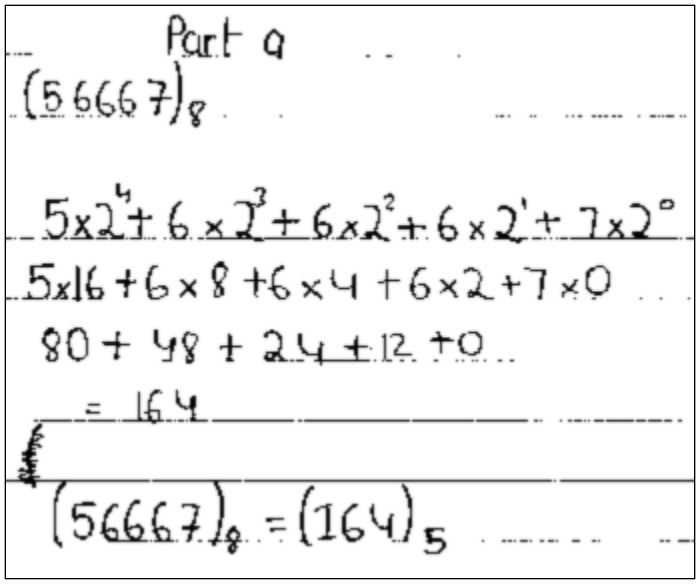
Example:



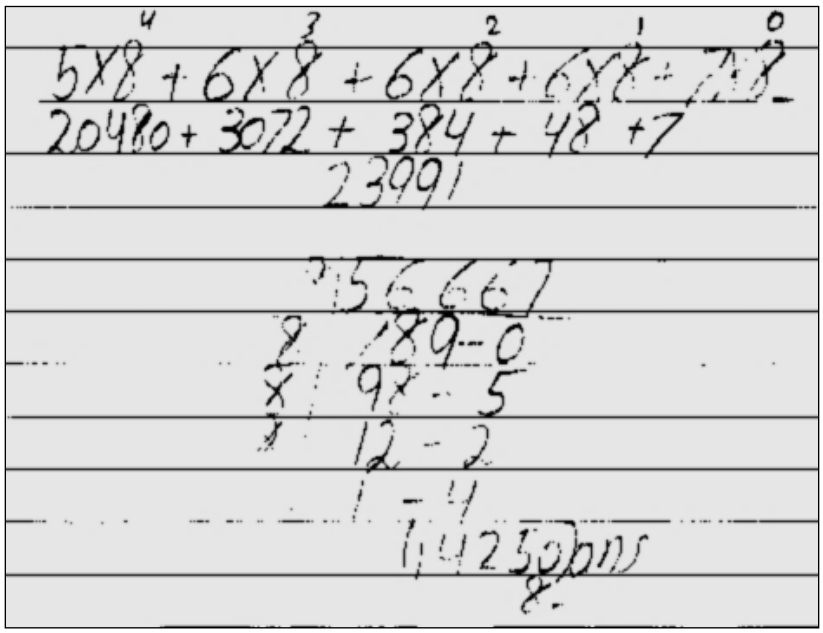
Weaker responses showed that the candidates had confusions in conversion of octal number to quinary number. Some of mistakes are as follows.

- Performed wrong conversion of octal numbers and treated the given numbers as a binary number
- Failed to follow the required sequence in conversion of numbers
- Converted octal number to decimal numbers and then converted it back to octal number incorrectly as cited in example 2.

Example 1:



Example 2:



Question 3b:

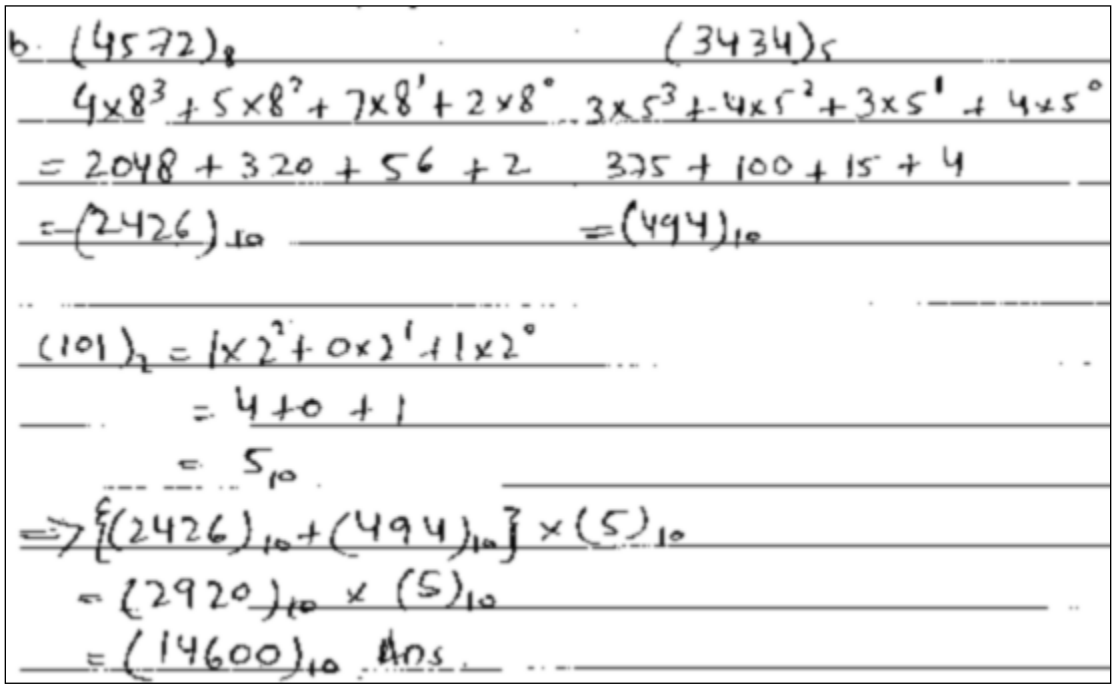
Perform the following operations and write your answer in decimal number (number in base 10).

$\{(4572)_8 + (3434)_5\} \times (101)_2$

(**Note**: Write all necessary steps)

Better responses indicated that the candidates were able to understand and apply their skills in the operations of addition and multiplication on different number systems. The candidates converted all the numbers to their decimal equivalents and then performed the required operation skilfully to get the answer.

Example:

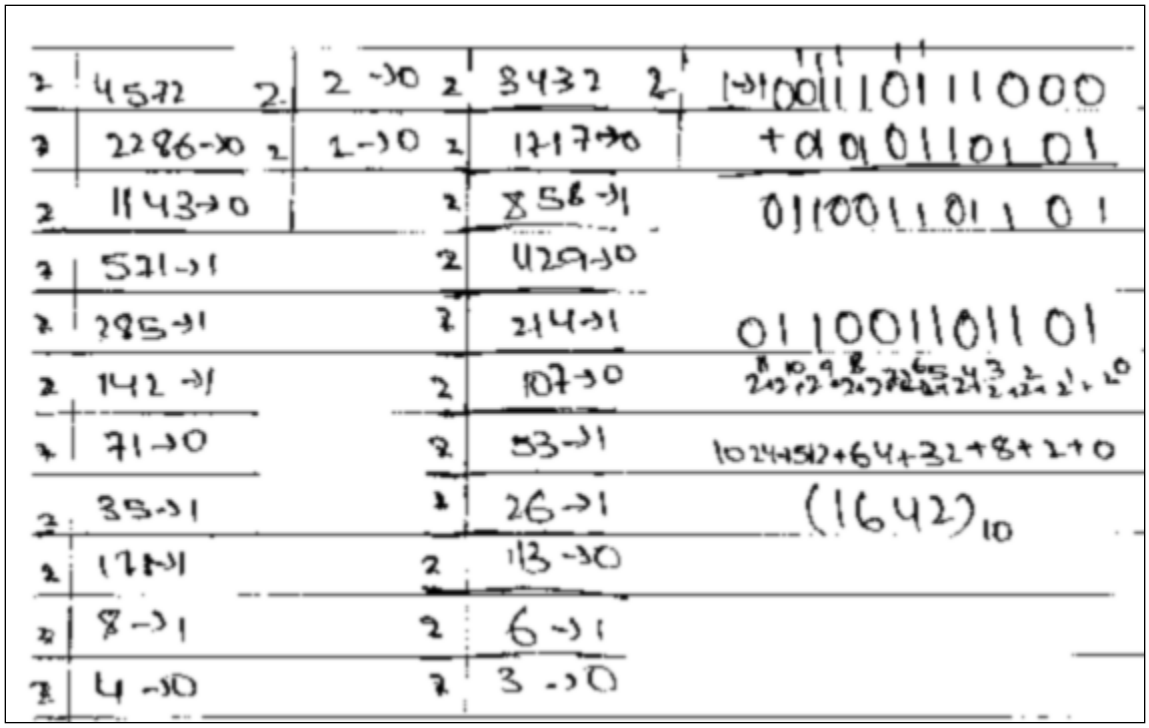


Weaker responses indicated that the candidates failed to perform the number operations on the given numbers correctly. They mainly failed to understand the importance of conversion of each number to same base or base 10.

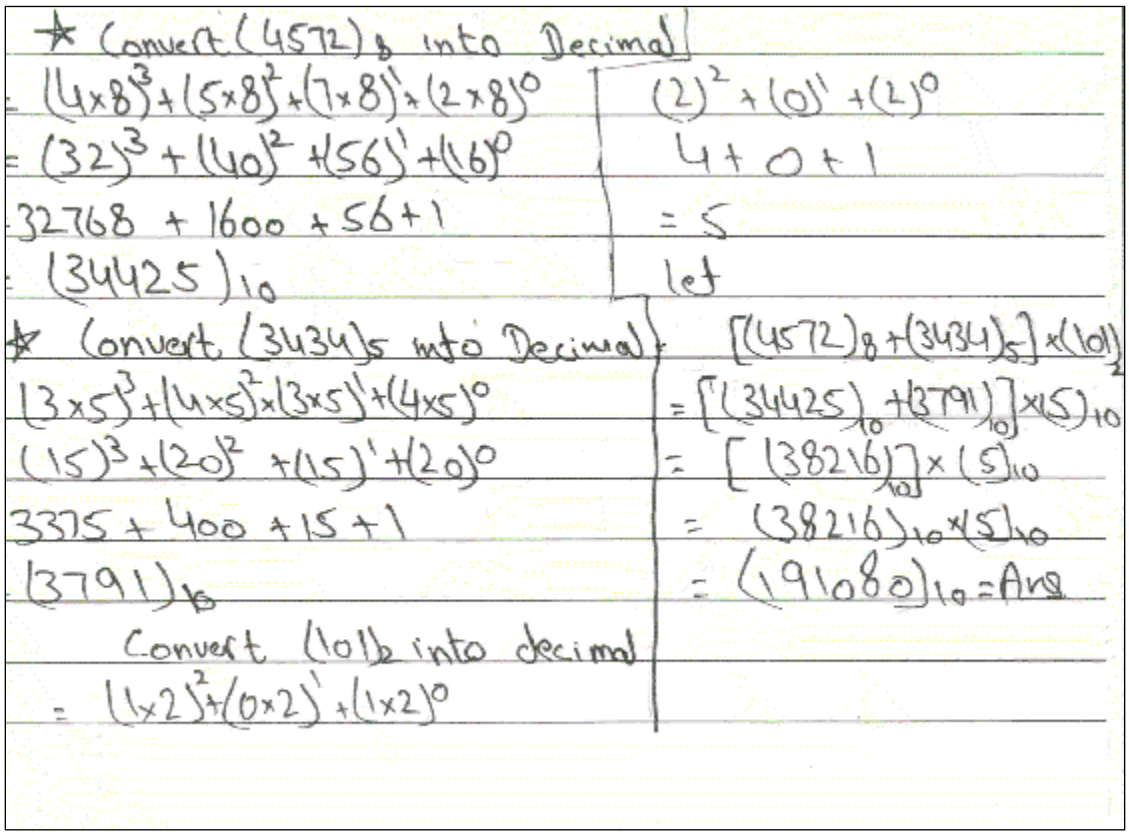
Secondly, they incorrectly converted number from different bases to base 10.

In Example 2, it is clear that candidates know the procedure of conversion but instead of writing as $(56667)_8 = 5\times8^4+6\times8^3+6\times8^2+6\times8^1+7\times8^0$ s/he wrote $(56667)_8 = (5\times8)^4+(6\times8)^3+(6\times8)^2+(6\times8)^1+(7\times8)^0$ which is incorrect way of conversion.

Example 1:



Example 2:



Question 4a:

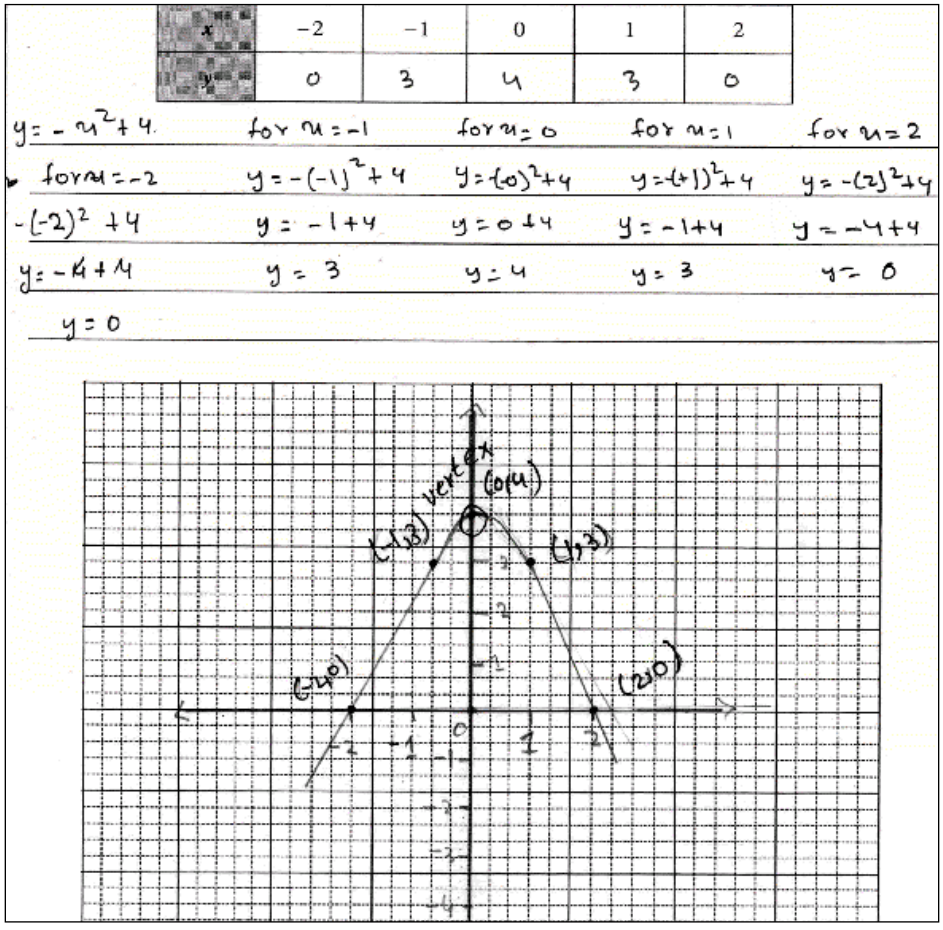
Fill the following table to sketch $f(x) = -x^2 + 4$ on the given graph paper and locate its vertex.

x	-2	-1	0	1	2
y					

This question offered a choice between part **a** and part **b**. Both parts were equally attempted by the candidates.

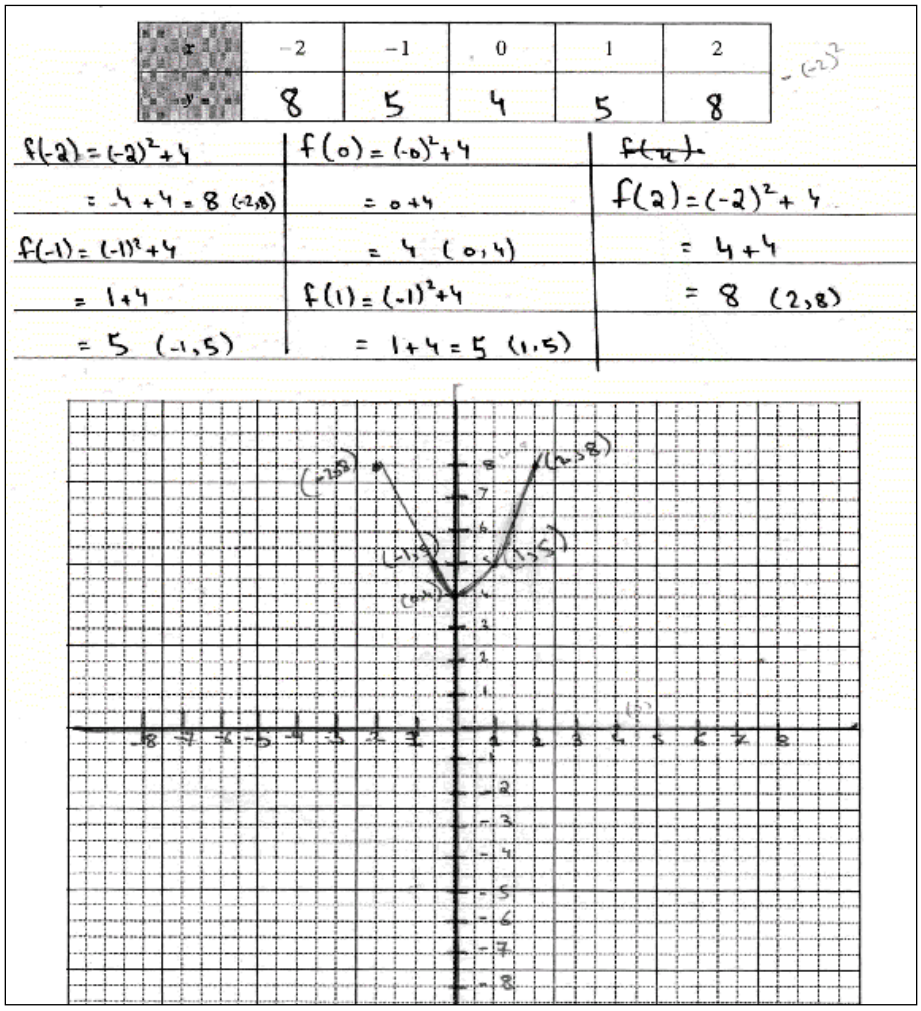
Better responses indicated that candidates correctly found the values of *x*. They choose the appropriate scales on the x-axis and y-axis and correctly located the points to obtain the required graph. They further wrote vertex of the parabola to fulfil the requirement of the question.

Example:

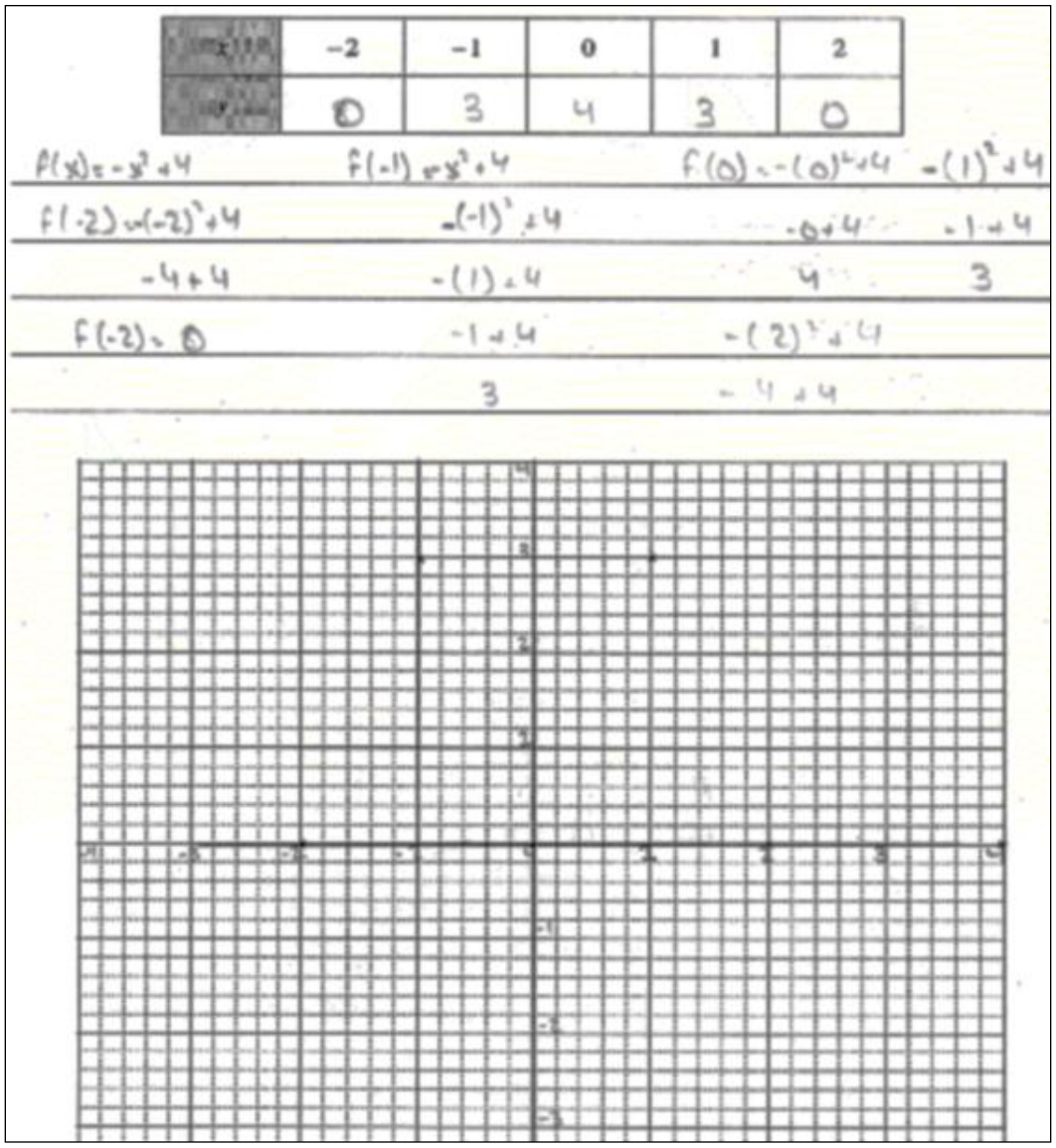


Weaker responses displayed that the candidates failed to find the values of x correctly or were unable to select proper scale to draw the required graph of the given function $f(x) = -x^2 + 4$.

Example 1:



Example 2:



x	−2	−1	0	1	2
y	0	3	4	3	0

$f(x) = -x^2 + 4$

$f(-2) = -(-2)^2 + 4$

$-4 + 4$

$f(-2) = 0$

$f(-1) = -x^2 + 4$

$-(-1)^2 + 4$

$-(1) + 4$

$-1 + 4$

3

$f(0) = -(0)^2 + 4$

$-0 + 4$

4

$-(2)^2 + 4$

$-4 + 4$

$-(1)^2 + 4$

$-1 + 4$

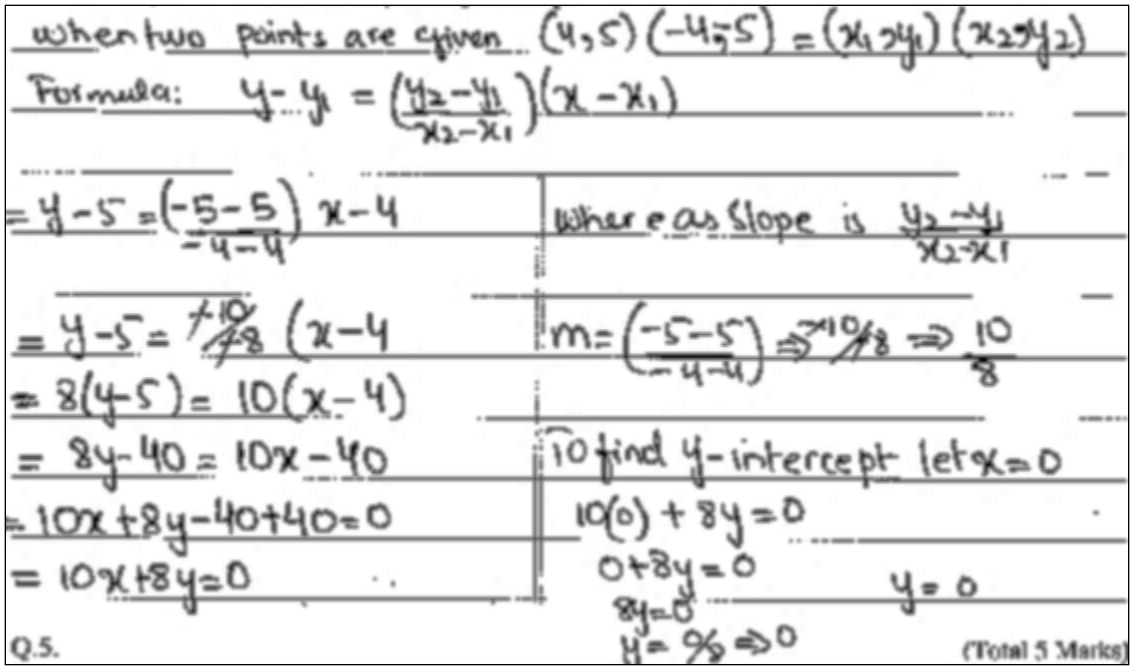
3

Question 4b:

Find the equation of the straight line passing through the points (4, 5) and $(-4, -5)$ in the form $ax + by + c = 0$. Find its slope and *y*-intercept.

Better responses selected the correct form of the equation and applied the form aptly to find the required equation. They had a clear concept of slope and *y*-intercept and able to find their values correctly.

Example:



Weaker responses displayed that candidates did not understand the question. They were confused among the different forms of straight line. These responses also indicated poor skills of algebraic operations. Some of the mistakes are recorded as follows.

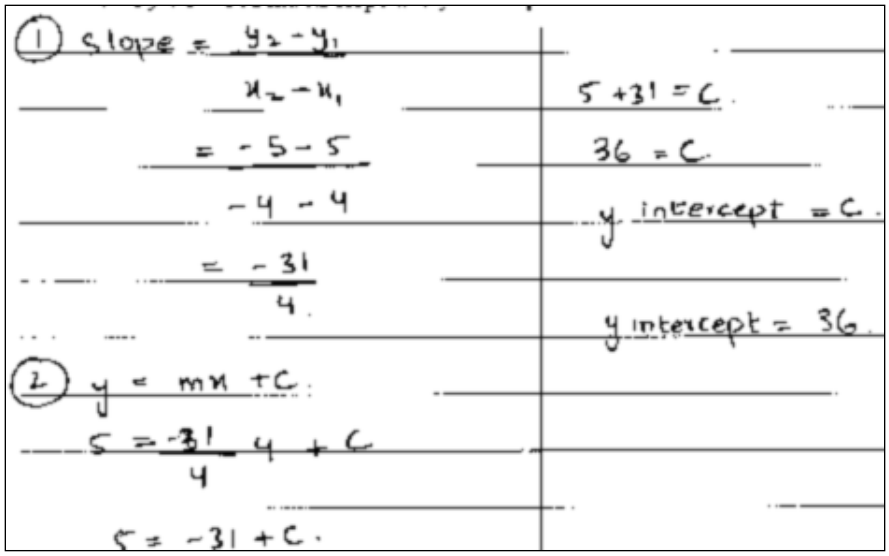
- For *y*-intercept, they substituted $y = 0$.
- They treated coefficient of *x* and *y* of the required equation $10x + 8y = 0$ as *x* and *y* intercepts.
- On substituting $y = 0$, in the equation $10x + 8y = 0$, they multiplied zero incorrectly as shown below.

  $10x + 8(0) = 0$

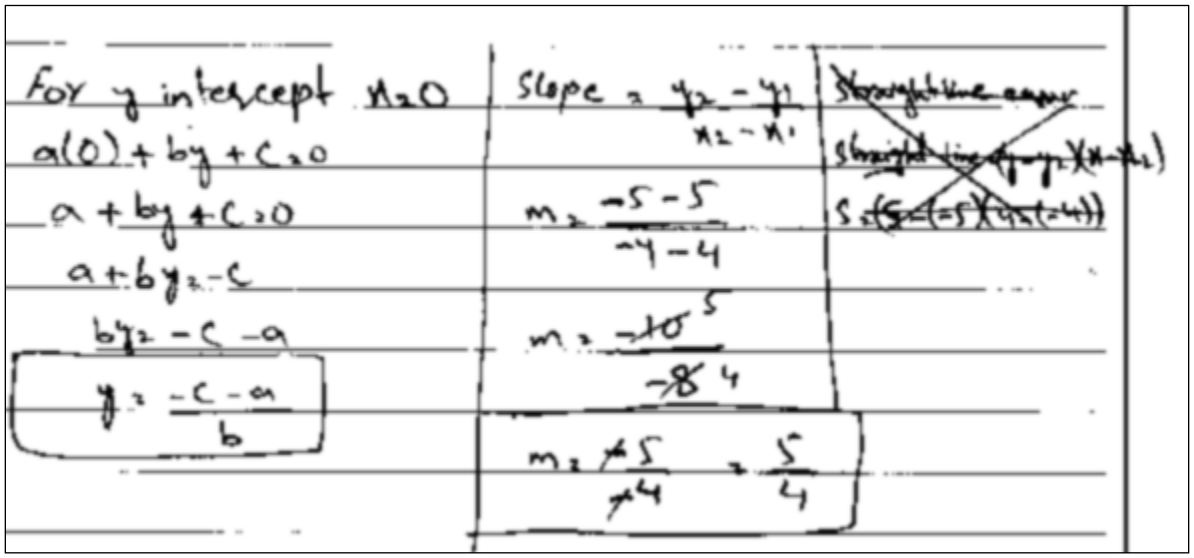
  $10x + 8 = 0$

- Wrong identification of values of x_1, x_2, y_1 and y_2 are also observed in few of the weaker responses

Example 1:



Example 2:



Question 5:

Eliminate k from the given equations.

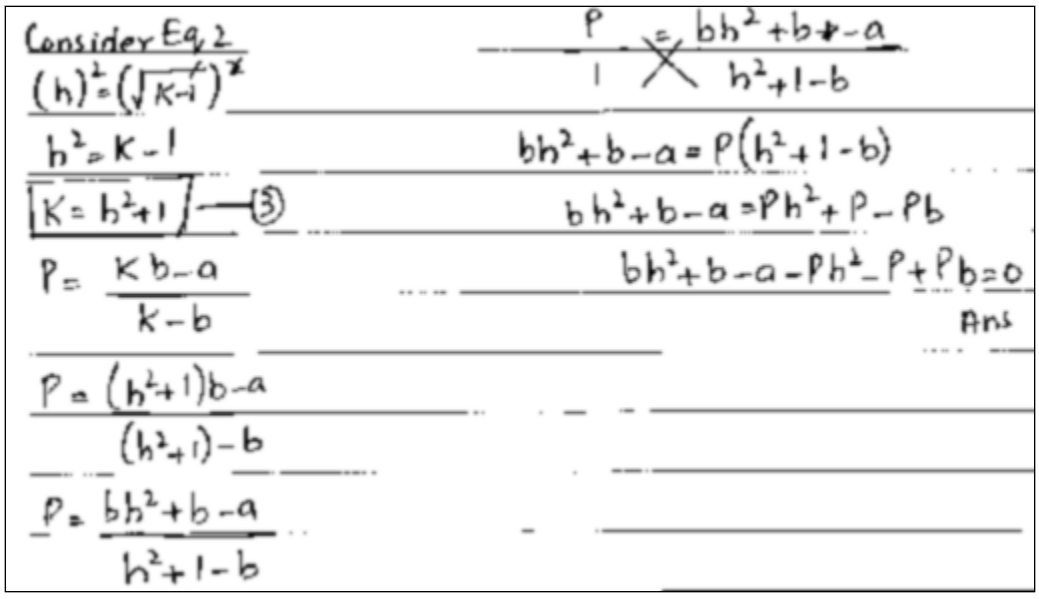
$$p = \frac{kb - a}{k - b}$$

$$h = \sqrt{k - 1}$$

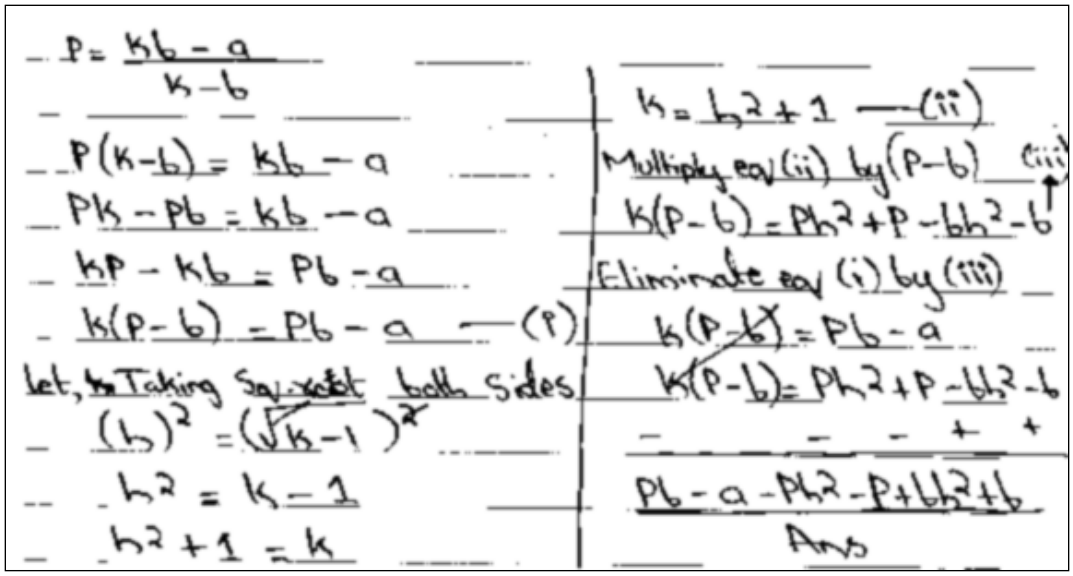
This was generally a well attempted question.

Better responses reported that the candidates were able to eliminate the k from the given equations. Mostly candidates got the value of the k from the equation by squaring both sides of $h = \sqrt{k-1}$ and substituted value in $p = \frac{kb-a}{k-b}$ to eliminate k followed by simplification of the result. In few responses, it is noted that candidates used the comparison method.

Example 1:



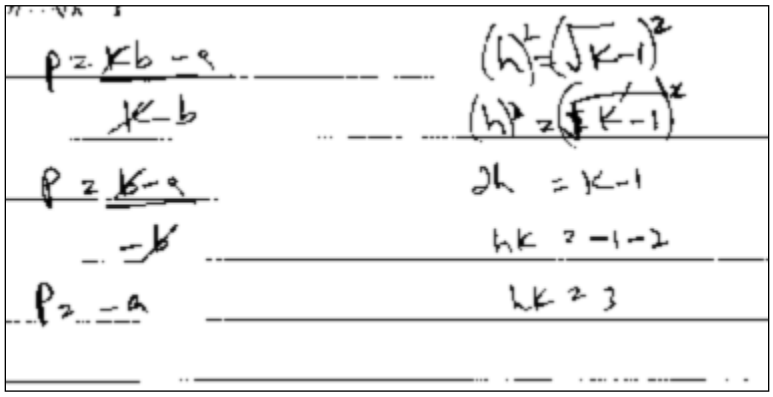
Example 2:



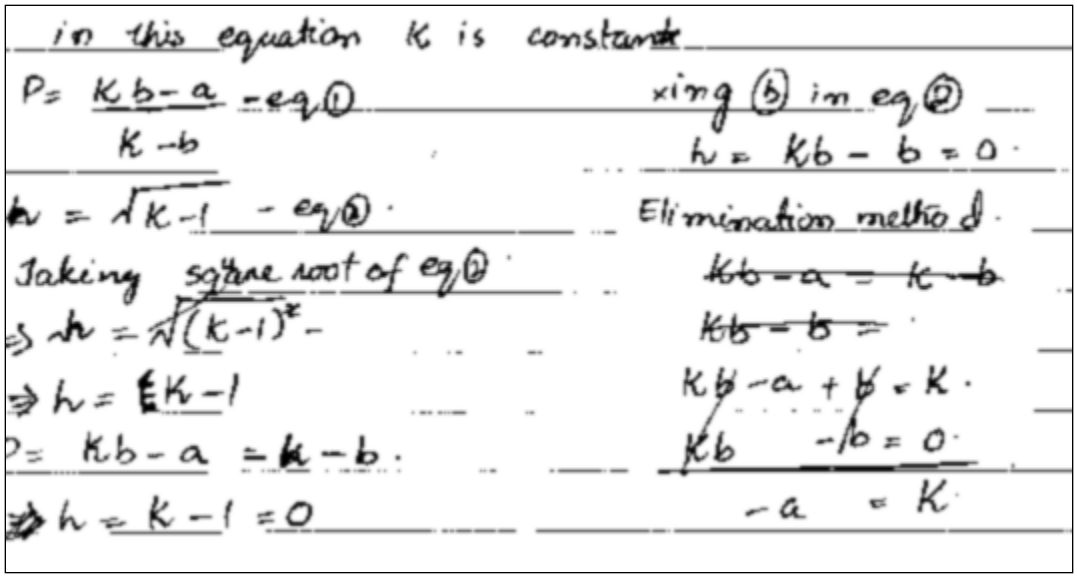
Weaker responses showed that the candidates were not equipped with the required skill to eliminate the required variable *h*. They made different mistakes and failed to eliminate *h*. Some of the mistakes are cited below.

- They wrongly cancelled *k* from the given equation $p = \frac{kb-a}{k-b}$ as $p = \frac{b-a}{1-b}$ or $p = \frac{b-a}{-b}$
- Some candidates made mistakes while taking square on both side of the equation; in few responses, it was noted that $2h = k-1$
- Some candidates found the value of *h* and did not proceed further.
- Some candidates unnecessarily compared the given equation with zero and started solving the equation

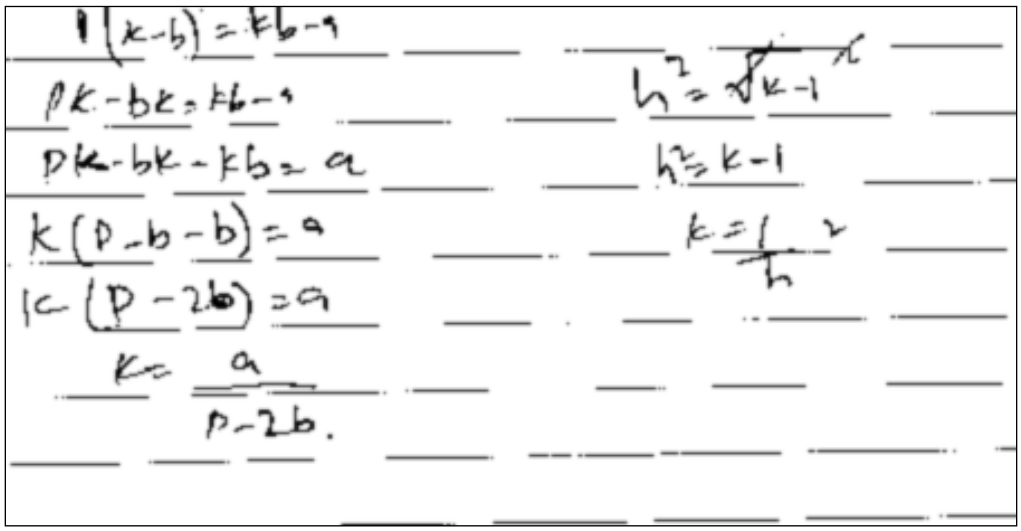
Example 1:



Example 2:



Example 3:



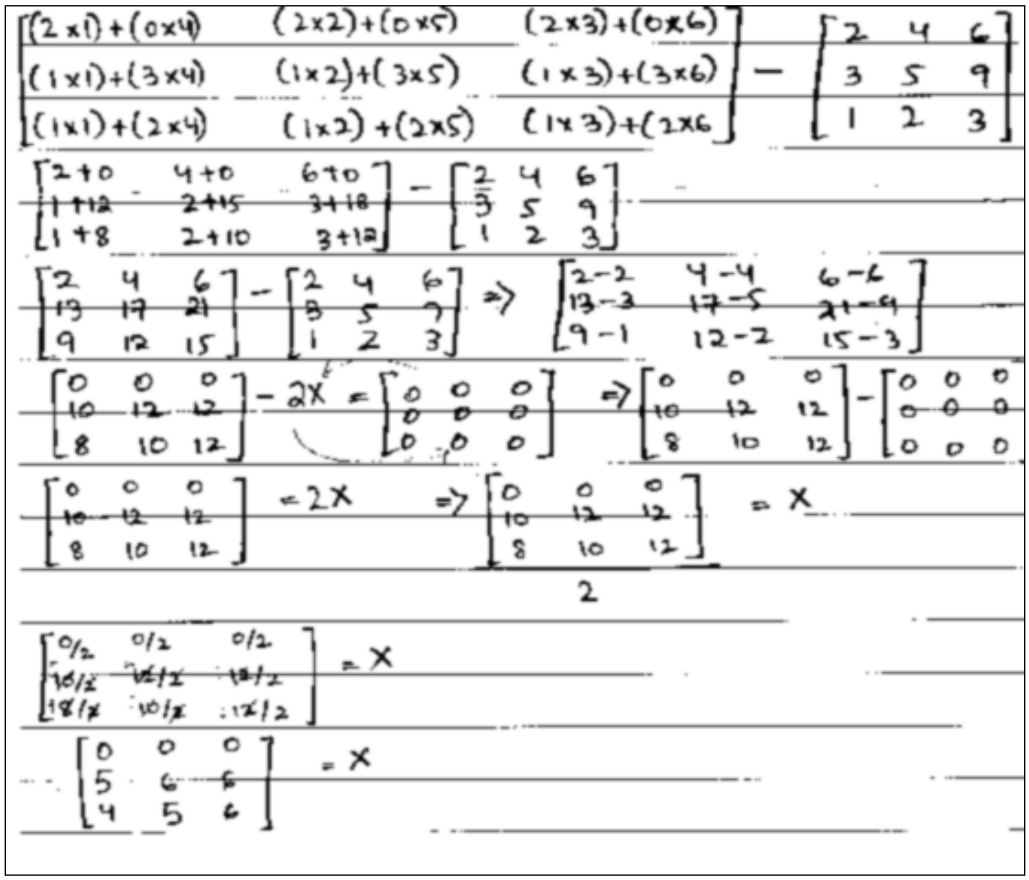
Question 6:

Solve the following matrix equation for the matrix *X*.

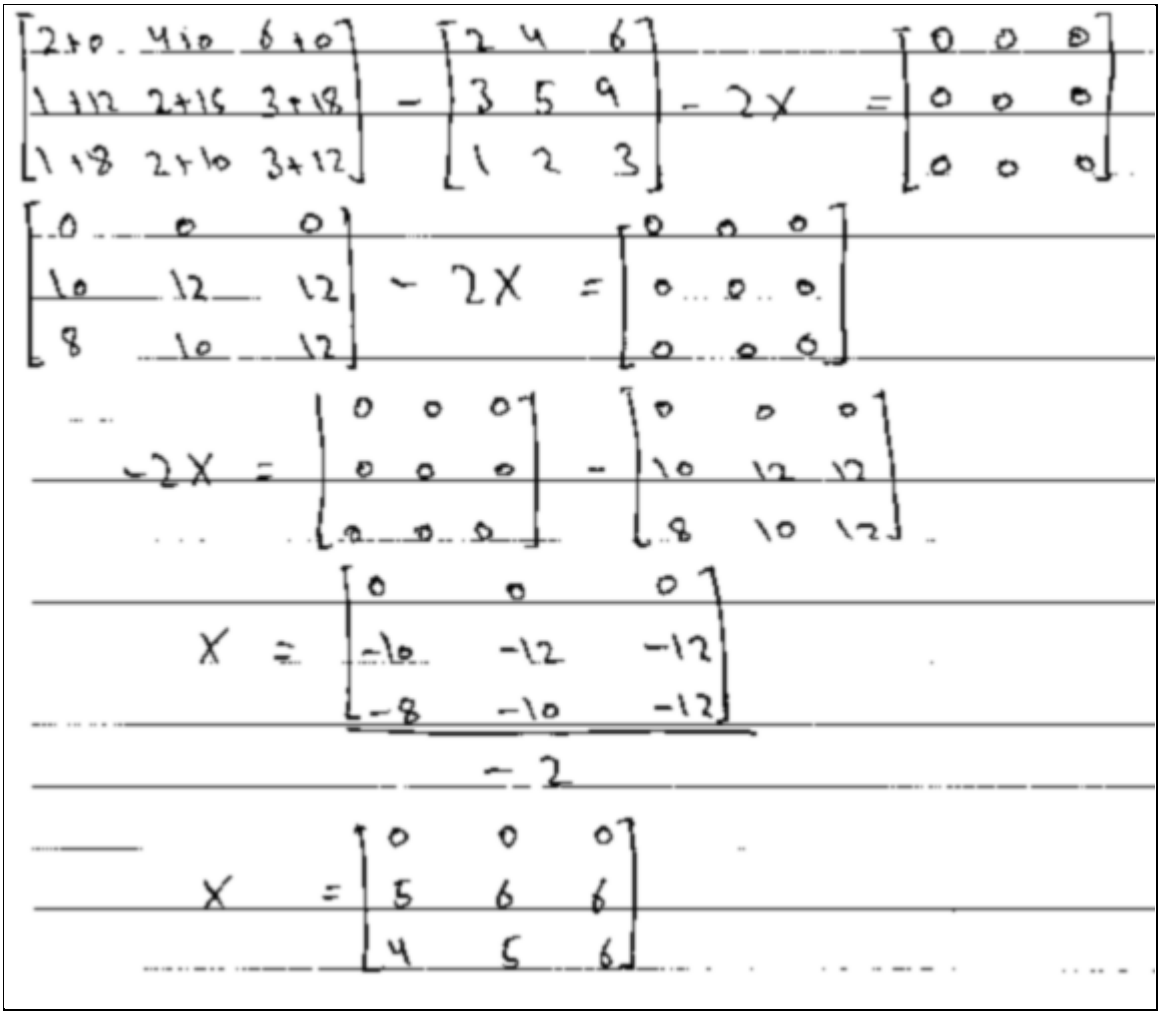
$$\begin{bmatrix}2 & 0\\1 & 3\\1 & 2\end{bmatrix}\times\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}-\begin{bmatrix}2 & 4 & 6\\3 & 5 & 9\\1 & 2 & 3\end{bmatrix}-2X=\begin{bmatrix}0 & 0 & 0\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix}$$

Better responses indicated that the candidates were well-versed in solving the problem based on the concept of matrices operations. Candidate correctly multiplied the matrices $\begin{bmatrix}2 & 0\\1 & 3\\1 & 2\end{bmatrix}\times\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}$. Finally, they subtracted and added the matrices correctly to get the value of Matrix *X*.

Example 1:



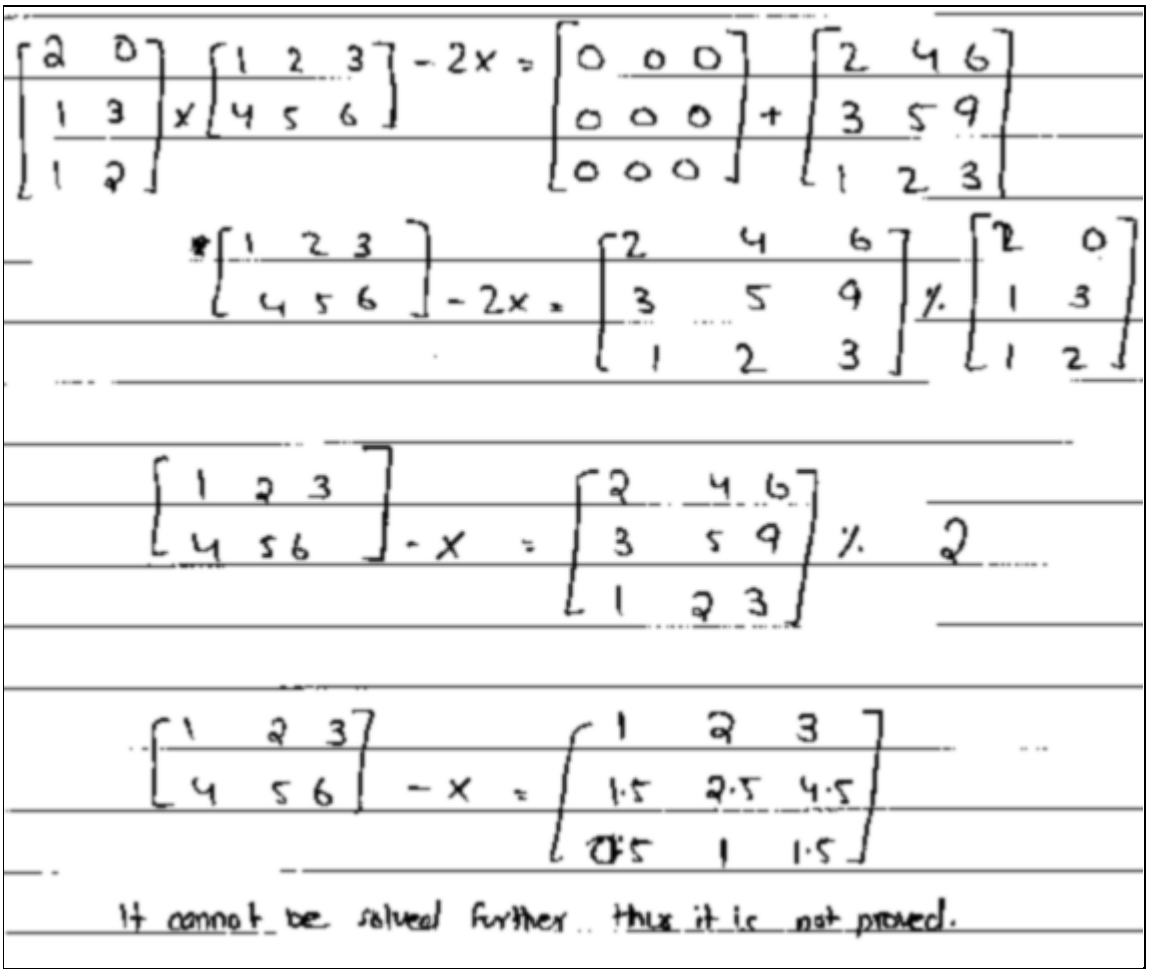
Example 2:



Weaker responses displayed that candidates failed to find the value of matrix *X*. Some mistakes noted are as follow.

- In some weaker responses, it was noted that candidates failed to multiply the given two matrices.
- Some candidates failed to follow the hierarchy of the matrices operations.
- In few cases, it was noted that candidates treated the given question to prove
- L.H.S =R.H.S and failed to do so as it was not the requirement of the question.
- Some candidates took transpose of the given matrix directly and then divided or multiplied it by the determinant.

Example 1:



Example 2:

